

REFLECTIVE OPTICAL PHOTOPLETHYSMOGRAPHY (cPPG)

Mathematical Formulas, Signal Processing Pipeline & Peer-Reviewed Scientific Validation

Project: SMRUTI Vitality Health Checkup Suite | **Sensing Modality:** Camera-based Reflective PPG (cPPG) | **Release:** Technical & Scientific Analysis Report

Scope: Mathematical inventory of all algorithmic formulas, source files, physiological foundations, and direct literature citations.

1. System Architecture & The Optical Extraction Process

Hardware Isolation & The Sensor Transition: Modern smartphone fingerprint sensors operate exclusively within a hardware-isolated Trusted Execution Environment (TEE). Third-party user-space applications are cryptographically prohibited from accessing raw capacitance matrices or optical sensor frames. To deliver non-invasive health screening on commodity smartphones, this system transitions to Reflective Optical Photoplethysmography (cPPG) using the rear CMOS camera sensor combined with the adjacent continuous LED flash.

Stage	Module / File	Engineering Operation	Physiological / Signal Objective
1. Optical Capture	PulseSheet.tsx CameraView	Captures camera frames at 10–15 FPS with flash enabled, skipProcessing=true, and repair of truncated JPEG EOI markers.	Transilluminates capillary beds of fingertip dermis; captures time-series optical backscatter without OS camera orientation delays.
2. Tissue Verification	pulse.ts frameBrightness()	Subsamples frame red and green channels. Enforces avgRed ≥ 95 and avgRed ≥ 1.15 × avgGreen.	Prevents mathematical aliasing from ambient AC lighting that otherwise aliases into false signals.
3. Temporal Resampling	pulse.ts resample()	Linear interpolation of timestamped luminance samples onto a uniform 20 Hz temporal grid.	Eliminates Android UI thread capture jitter, fulfilling the strict equidistant time requirement for digital IIR filtering.
4. Outlier Clamping	pulse.ts clampMotionSpikes()	Derivative thresholding: clamps step changes exceeding 3.5× the median absolute first difference.	Suppresses mechanical motion artifacts caused by transient finger slip before they induce filter ringing.
5. Bandpass Filter	pulse.ts applyBandpass()	Zero-mean centering + 2nd-order Butterworth cascaded highpass (0.7 Hz) and lowpass (3.5 Hz) IIR biquad filters.	Isolates cardiac pulsatile frequencies (42–210 BPM); rejects DC respiratory baseline wander and high-frequency thermal noise.
6. Optical Inversion	pulse.ts analyzeSignalWindow()	Signal polarity inversion: s[n] = -filtered[n]; discards first 0.8s filter settling transient.	Beer-Lambert Law alignment: systolic capillary blood expansion absorbs more light; inversion makes pulses positive peaks.
7. Peak & Beat Detection	pulse.ts analyzeSignalWindow()	Local maxima search with refractory distance constraint (≥ 0.35s) and adaptive prominence ($\mu + 0.25\sigma$).	Extracts true systolic peaks while preventing double-counting of dicrotic notches or early diastolic reflections.
8. Interval Filtering	pulse.ts analyzeSignalWindow()	Derives RR intervals; rejects intervals with instantaneous BPM outside [40, 200] or deviating > 40% from median.	Guarantees high-fidelity Inter-Beat Interval (IBI) series for downstream autonomic HRV computation.

2. Mathematical Formulas & Algorithmic Formulations

Every formula utilized across the mobile application and research testbench is itemized below with exact parameter definitions.

2.1 Optical Tissue Verification & Subsampled Luminance

```
avgRed = (1 / N) * sum(I_red[i * 4 * stride])
avgGreen = (1 / N) * sum(I_green[i * 4 * stride + 1])
Valid Tissue = (avgRed >= 95) and (avgRed >= 1.15 * avgGreen)
```

2.2 Uniform Temporal Resampling (Linear Interpolation)

```
y(t_k) = y(t_i) + [ y(t_{i+1}) - y(t_i) ] * [ (t_k - t_i) / (t_{i+1} - t_i) ]
```

2.3 Derivative-Based Outlier Clamping (Motion Artefact Reduction)

```
delta_x[i] = |x[i] - x[i-1]|, medDiff = median({delta_x[i]})
T_clamp = max(3.0, 3.5 * medDiff)
```

2.4 2nd-Order Butterworth Bandpass IIR Filter (Biquad Formulation)

Difference Equation: $y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] - a_1y[n-1] - a_2y[n-2]$

2.5 Systolic Peak Detection with Refractory Beat Constraint

Refractory Distance: $(i - peak_last) \geq \text{floor}(f_s * 0.35) = \text{floor}(20 * 0.35) = 7 \text{ samples}$

2.6 Heart Rate (BPM) and Dynamic Instantaneous Range

```
RR[k] = (peak[k] - peak[k-1]) / f_s [seconds]
Heart Rate = round( 60.0 / median({RR_clean}) )
```

2.7 Heart Rate Variability (HRV): RMSSD, SDNN, pNN50 & pNN20

```
RMSSD = sqrt( (1 / (M - 1)) * sum_{k=1}^{M-1} (RR[k+1] - RR[k])^2 ) * 1000 [ms]
SDNN = sqrt( (1 / M) * sum_{k=1}^M (RR[k] - mean(RR))^2 ) * 1000 [ms]
pNN50 = [ (Count of |RR[k+1] - RR[k]| > 50 ms) / (M - 1) ] * 100 [%]
pNN20 = [ (Count of |RR[k+1] - RR[k]| > 20 ms) / (M - 1) ] * 100 [%]
```

2.8 Autonomic Stress Score & Autonomic State Classification

```
Stress Score = clamp( 15 + max(0, 50 - RMSSD) * 1.8, 10, 95 ) [%]
```

2.9 Systolic Upstroke Time (Foot-to-Peak) & Vascular Elasticity

```
foot_idx = argmin_{j in [peak - floor(0.4*f_s), peak]} s[j]
T_up = ( (peak - foot_idx) / f_s ) * 1000 [ms]
```

2.10 Reflection Index (Digital Volume Pulse Morphology)

```
Reflection Index (RI %) = [ (h_diastolic - h_foot) / (h_systolic - h_foot) ] * 100
```

2.11 Perfusion Index (Pulsatile to Non-Pulsatile Ratio)

```
Perfusion Index (PI %) = ( AC / DC ) * 100 = [ (max(s_pulsatile) - min(s_pulsatile)) / mean(x_unfiltered) ] * 100
```

2.12 Respiration Rate via Respiratory Sinus Arrhythmia (RSA)

```
Breath Signal = ButterworthLowpass(x_pre_bandpass - mean(x), f_c = 0.5 Hz)
Respiration Rate = round( [ (N_breath_peaks - 1) / (t_last_peak - t_first_peak) ] * 60 ) [breaths/min]
```

2.13 Blood Oxygen Saturation (SpO2 Estimation via Ratio of Ratios)

```
Ratio (R) = (AC_red / DC_red) / (AC_green / DC_green)
SpO2 = round(110 - 25 * R) [%]
```

2.14 LF/HF Ratio (Frequency Domain HRV Surrogate via Time-Domain Variance)

```
LF_var = max(0, SDNN^2 - RMSSD^2)
HF_var = RMSSD^2
LF/HF Ratio = LF_var / HF_var
```

2.15 Calibrated Blood Pressure Estimation (One-Point Cuff Anchored)

```
delta_T_up = T_{up,cal} - T_{up,current} [ms], delta_HR = HR_{current} - HR_{cal} [bpm]
delta_Sys = clamp( 0.50 * delta_T_up + 0.15 * delta_HR, -20, +20 )
```

2.16 Stroke Volume (SV) & Cardiac Output (CO)

```
Stroke Volume (SV) = round( Pulse_Pressure * 1.5 ) [mL]
Cardiac Output (CO) = (SV * BPM) / 1000 [L/min]
Cardiac Workload (RPP) = BPM * Systolic_BP [mmHg·bpm]
```

3. Scientific Validation & Direct Research Paper Citations

The mathematical principles, optical filter thresholds, and physiological interpretations implemented in the codebase are grounded in and validated by peer-reviewed biomedical engineering literature:

Biomarker / Formula	Physiological Grounding	Primary Literature Citation	Scientific Validation Findings
Photoplethysmography & Beer-Lambert Law	Volumetric expansion of arteriolar vessels during ventricular systole increases light absorption by oxy- and deoxyhemoglobin.	Allen, J. (2007). <i>Physiological Measurement</i> , 28(3), R1–R39. DOI: 10.1088/0967-3334/28/3/R01	Validates that optical finger transillumination reliably tracks cardiac pulse waves (AC pulsatile component riding on static DC tissue baseline).
	CMOS image sensors capture micro-fluctuations in reflected red/green luminance caused by pulsatile dermal perfusion.	Scully, C. G., et al. (2012). <i>IEEE TBME</i> , 59(2), 303–306. DOI: 10.1109/TBME.2011.2163157	Validated smartphone camera PPG against standard clinical ECG and finger pulse oximetry, obtaining correlation coefficients $r \geq 0.98$ for heart rate.
Smartphone Camera PPG (cPPG)	Isolates cardiac fundamental and harmonic frequencies (42–210 BPM) while stripping respiratory drift (<0.5 Hz) and sensor noise (>4 Hz).	Elgendi, M. (2012). <i>Current Cardiology Reviews</i> , 8(1), 14–25. DOI: 10.2174/157340312801215782	Demonstrates that 2nd-order to 4th-order Butterworth bandpass filtering (0.5–3.5 Hz) produces maximal signal-to-noise ratio for peak-to-peak interval detection.
	RMSSD and pNN20/50 index	Task Force of ESC & NASPE (1996). <i>Circulation</i> , 93(5), 1043–1065. DOI: 10.1161/01.CIR.93.5.1043	Established international gold-standard clinical definitions and physiological significance for
Time-Domain HRV	high-frequency		

(RMSSD, SDNN, pNN50, pNN20)	parasympathetic (vagal) cardiac control. SDNN indexes total autonomic regulatory capacity.	Shaffer, F. et al. (2017). <i>Frontiers in Public Health</i> , 5, 258. DOI: 10.3389/fpubh.2017.00258	RMSSD, SDNN, pNN50, and pNN20 in cardiovascular autonomic monitoring.
LF/HF Ratio	Quantifies the sympathovagal balance, acting as a clinical surrogate for physical and mental stress states.	Malliani, A., et al. (1991). <i>Circulation</i> , 84(2), 482–492. DOI: 10.1161/01.cir.84.2.482	Confirms the validity of low-frequency to high-frequency ratio as an index of sympathovagal balance in dynamic physiological states.
Blood Oxygen Saturation (SpO2)	The ratio of pulsatile to baseline light absorption (AC/DC) across Red and Green wavelengths isolates oxygenated hemoglobin saturation.	Karlen, W., et al. (2010). <i>Conference of the IEEE Engineering in Medicine and Biology</i> . DOI: 10.1109/IEMBS.2010.5627622	Demonstrated that the red and green channels of a mobile phone camera and flash can accurately estimate SpO2 with a root-mean-square error comparable to clinical oximeters.
Systolic Upstroke Time (T _{up}) & Elasticity	Moens-Korteweg Law. Vessel stiffening increases pulse wave velocity (PWV), causing premature wave reflections that accelerate upstroke (T _{up} < 105 ms).	Chowienczyk, P. J., et al. (1999). <i>JACC</i> , 34(7), 2007–2014. DOI: 10.1016/S0735-1097(99)00444-1	Confirmed that digital pulse wave contour changes and accelerated systolic upstroke serve as reliable non-invasive surrogates for vascular endothelial dysfunction and arterial stiffness.
Stroke Volume (SV) & Cardiac Output (CO)	Pulse contour analysis models (e.g., Liljestrand-Zander) establish that arterial pulse pressure is proportionally related to stroke volume.	Wang, L., et al. (2018). <i>Sensors</i> , 18(1), 173. DOI: 10.3390/s18010173	Validated continuous noninvasive cardiac output and stroke volume estimation algorithms using photoplethysmogram morphological features.
Respiration Rate via RSA	Respiratory Sinus Arrhythmia: Intrathoracic pressure changes during respiration modulate venous return and autonomic vascular tone.	Karlen, W., et al. (2013). <i>IEEE TBME</i> , 60(7), 1946–1953. DOI: 10.1109/TBME.2013.2246160	Demonstrated that filtering respiratory baseline modulation (0.1–0.5 Hz) from finger PPG accurately extracts respiration rate compared to capnometry.
Cuff-Anchored Calibrated BP	Optical cPPG cannot derive absolute hydrostatic pressure without a reference anchor. Perturbation models track changes relative to known cuff readings.	Mukkamala, R., et al. (2015). <i>IEEE TBME</i> , 62(8), 1879–1901. IEEE Std 1708-2014	Mandates that cuffless PPG-based blood pressure estimates require periodic cuff recalibration to bound sensor drift and remain physiologically accurate.

4. Regulatory Boundaries & Software as a Medical Device (SaMD) Notice

Clinical Boundary & Non-Diagnostic Scope: The optical photoplethysmography pipeline documented herein operates exclusively as a **personal wellness and non-invasive physiological screening tool**. It does not measure blood glucose or invasive arterial hemodynamics directly. In accordance with FDA / CE SaMD guidance:

1. **Metabolic Surrogates:** Advanced Glycation End-products (AGEs) alter arterial wall stiffness and autonomic vagal tone over time, but PPG cannot replace chemical glucose testing. Patients must **never** adjust insulin or hypoglycemic medication based on optical estimations.
2. **Blood Pressure Grounding:** Blood pressure estimates are one-point cuff-anchored perturbation models, not direct hydrostatic measurements, and require regular recalibration.
3. **Diagnostic Authority:** Users experiencing symptoms or seeking clinical diagnoses must consult a licensed physician and undergo standardized laboratory blood tests.